Cessna 310R (C310R)

2× Continental IO-520-M, 285 hp · 6 seats · Typical 310R (1975–1981, 5,500 lb) POH procedures; counter-weighted feathering propellers, 100 A alternators, main (tip) tanks with optional auxiliary wing and wing locker tanks.

V-speeds & limits

Vmca	**75 KIAS (red radial)**
Vsse	**83–88 KIAS (intentional one-engine-inop; verify in POH)**
Vr	**approx 90 KIAS (liftoff, normal takeoff)**
Vx	**84 KIAS**
Vxse	**93 KIAS**
Vyse	**106 KIAS (blue line)**
Vy	**107 KIAS**
Vfe	**160 KIAS (15°) / 140 KIAS (full 35°)**
Vle / Vlo	**140 KIAS**
Va	**148 KIAS (5,500 lb)**
Vno	**181 KIAS**
Vne	**223 KIAS**
Vs / Vso	**approx 79 / 72 KIAS (5,500 lb)**
Vref	**90–95 KIAS final (full flaps, max weight); reduce approx 1 KIAS per 100 lb below max**

Normal procedures

Before Start

Preflight inspection	**COMPLETE (fuel drains, tip tanks, nose / wing locker baggage doors latched)**
Passenger briefing	**COMPLETE (belts, cabin door, emergency exit window, no smoking)**
Seats, belts, shoulder harnesses	**ADJUST & LOCK**
Cabin door	**CLOSED & LATCHED (handle flush)**
Parking brake	**SET**
Avionics power switch	**OFF**
Landing gear handle	**DOWN**
Throttles / propellers / mixtures	**IDENTIFY – throttles closed, props high RPM, mixtures idle cutoff**
Fuel selectors	**MAIN (left – LEFT MAIN, right – RIGHT MAIN; feel for detents)**
Fuel quantity	**CHECK mains and aux (gauge selector switch), locker tanks if fitted**
Cowl flaps	**OPEN**
Alternate static source	**OFF (normal)**
Battery switch	**ON**
Alternator switches	**ON**
Landing gear lights	**GREEN (down); gear unsafe light OUT**
Annunciator / warning lights	**PRESS TO TEST**
Beacon	**ON**

Engine Start

Propeller area	**CLEAR (both sides)**
Magneto switches	**ON (both)**
Mixture	**FULL RICH**
Propeller	**HIGH RPM**
Throttle	**1/2 OPEN**
Auxiliary fuel pump	**HIGH until fuel flow peaks (3–5 sec), then OFF (LOW in hot weather)**
Throttle	**1/4 OPEN**

▶ Engine hot or flooded?

No:

Yes:

Mixture	**IDLE CUTOFF**
Throttle	**FULL OPEN**
Mixture	**ADVANCE to FULL RICH as engine fires, throttle to idle**
Starter switch	**ENGAGE (release when engine fires)**
Throttle	**800–1000 RPM**
Oil pressure	**10 psi within 30 sec, rising to 30–60 psi**
Second engine	**REPEAT start sequence**
Alternators	**CHECK both charging (ammeters / voltmeter approx 28 V, ALT lights out)**
External power	**DISCONNECTED (if used)**
Avionics power switch	**ON**
Engine instruments	**CHECK both (fuel flow steady)**

Before Taxi

Brakes	**CHECK on initial roll**
Flight instruments	**CHECK & SET (altimeters, HSI / DG, turn coordinator during turns)**
Avionics / GPS	**SET**
Autopilot / electric trim	**CHECK & OFF**
Trims (elevator, rudder, aileron)	**SET for takeoff**
Fuel selectors	**CHECK MAIN both**
Cowl flaps	**OPEN**

Before Takeoff

Parking brake	**SET**
Flight controls	**FREE & CORRECT**
Fuel selectors	**MAIN both (takeoff and landing on main tanks only)**
Mixtures	**FULL RICH (lean per fuel flow gauge above approx 5,000 ft elevation)**
Throttles	**1700 RPM (one engine at a time)**
Magnetos	**CHECK L & R each engine (150 RPM max drop, 50 RPM max difference)**
Propellers	**CYCLE 2–3 times (RPM drops, then returns)**
Propellers – feather check	**PULL to FEATHER momentarily at 1500–1700 RPM – RPM drops (approx 200–300 RPM); return before RPM falls below 1000**
Engine instruments	**CHECK both (oil pressure 30–60 psi, oil temp green, fuel flow steady, CHT green)**
Alternators	**CHECK both charging (load with pitot heat / lights, 100 A each)**
Gyro suction	**CHECK in green, both pump indicators clear**
Throttles	**IDLE check (600–650 RPM), then 1000 RPM**
Auxiliary fuel pumps	**LOW for takeoff (OFF if operator checklist specifies; HIGH only with a failed engine-driven pump)**
Cowl flaps	**OPEN**
Propellers	**HIGH RPM**
Flaps	**UP (15° short field) – check indicator**
Trims	**SET takeoff range**
Landing gear handle	**DOWN – IDENTIFY**
Pitot heat / de-ice	**AS REQUIRED**
Lights	**AS REQUIRED**
Transponder	**ALT**
Cabin door / windows	**LATCHED**
Takeoff briefing	**COMPLETE (Vmca 75 / abort below 90 / Vyse 106; engine-out plan)**

After Takeoff / Climb

Airspeed	**ACCELERATE to 106 KIAS (Vyse) minimum before gear retraction**
Landing gear	**UP (positive rate, no usable runway, below 140 KIAS)**
Gear lights	**OUT (gear up and locked)**
Flaps	**UP (if used)**
Climb power	**FULL THROTTLE / 2700 RPM to safe altitude, then 25 in Hg / 2500 RPM cruise climb**
Airspeed	**107 KIAS (Vy) / 120 KIAS cruise climb**
Mixtures	**LEAN per fuel flow gauge climb schedule above approx 5,000 ft**
Cowl flaps	**OPEN (CHT / oil temp in green)**
Auxiliary fuel pumps	**OFF (LOW if fuel flow fluctuates)**
Fuel selectors	**MAIN (aux tanks not for climb)**
Engine instruments	**CHECK both**

Cruise

Cruise power	**SET (2300–2450 RPM, 22–24 in Hg typical; 65–75%)**
Mixtures	**LEAN per fuel flow gauge cruise marks (approx 13–15 GPH per engine) or 50°F rich of peak EGT**
Cowl flaps	**CLOSED (as required for CHT)**
Propellers	**SYNCHRONISE**
Fuel management	**MAIN tanks for the first 60 min (20 gal aux) / 90 min (31.5 gal aux) of flight**
Auxiliary tanks	**SELECT AUX only in level cruise, after the mains have room; return to MAIN before descent with aux nearly empty**

Fuel-injection return (vapour) flow goes to the MAIN tank on that side regardless of selector position. Using the aux tanks before the mains have room vents fuel overboard. Aux tanks are not for takeoff, climb, landing or crossfeed.

Wing locker transfer (if fitted)	**PUMP ON when mains have room (approx 20 gal per side); OFF when amber transfer light illuminates (tank empty)**
Fuel quantity / flow	**CHECK both sides against plan**
Fuel selectors	**CROSSFEED only for one-engine-inop balance in level flight (mains only)**
Engine instruments	**CHECK both (oil temp 240°F max, CHT 460°F max)**
Altimeter / heading	**CHECK**

Descent

Altimeters	**SET**
Fuel selectors	**MAIN both (sufficient fuel in mains for approach, go-around and reserve)**
Wing locker transfer pumps	**OFF**
Mixtures	**ENRICH progressively**
Power	**REDUCE gradually (avoid rapid cooling)**
Cowl flaps	**CLOSED**
Airspeed	**PLAN below 160 KIAS for approach flaps, 140 KIAS for gear**
Approach briefing	**COMPLETE (incl. single-engine go-around decision point)**
Seat belts / harnesses	**SECURE**

Before Landing

Fuel selectors	**MAIN both**
Auxiliary fuel pumps	**LOW**
Mixtures	**FULL RICH**
Cowl flaps	**AS REQUIRED**
Airspeed	**BELOW 140 KIAS**
Landing gear	**DOWN – GREEN light, unsafe light out (identify gear handle)**
Flaps	**15° below 160 KIAS; FULL (35°) on final below 140 KIAS**
Propellers	**HIGH RPM (on final)**
Airspeed	**90–95 KIAS final (full flaps)**
Landing lights	**ON**
Landing gear	**GREEN – CONFIRMED on final**

After Landing

Flaps	**UP (identify flap handle before moving)**
Cowl flaps	**OPEN**
Auxiliary fuel pumps	**OFF**
Pitot heat / de-ice	**OFF**
Landing lights / strobes	**AS REQUIRED**
Transponder	**STBY / as required**
Trims	**SET takeoff**

Shutdown

Parking brake	**SET**
Throttles	**1000 RPM (cool-down)**
Avionics power switch	**OFF**
Electrical equipment / lights	**OFF**
Mixtures	**IDLE CUTOFF**
Magneto switches	**OFF (after propellers stop)**
Battery & alternator switches	**OFF**
Fuel selectors	**MAIN (leave on mains; OFF only for maintenance)**
Control lock	**INSTALL**
Chocks / tiedowns	**AS REQUIRED**

Abnormal procedures

Alternator Failure / Low Voltage

ALT warning light, one ammeter zero or voltage below 24 V

Affected alternator switch	**OFF, then ON (reset over-voltage relay)**
Alternator field circuit breaker	**CHECK / RESET once**

▶ Alternator restored?

Yes:

Ammeters / voltmeter	**MONITOR both**

No – one alternator failed:

Affected alternator switch	**OFF**
Electrical load	**REDUCE to within remaining alternator (100 A)**
Deferred: One alternator inoperative – electrical load reduced, land as soon as practical.	**NOTED**

No – both failed:

Both alternator switches	**OFF**
Non-essential electrical load	**REDUCE (battery only, limited endurance)**
Landing gear	**PLAN early extension while battery power remains (hand crank if needed)**
Deferred: Both alternators failed – battery only; extend gear early, flaps may be unavailable, land as soon as possible.	**NOTED**

Low Oil Pressure (one engine)

Oil pressure below 30 psi in flight

Oil pressure / oil temperature	**CHECK and cross-check the affected engine**

▶ Oil temperature rising or pressure below 10 psi?

Yes – genuine:

Affected engine power	**REDUCE to minimum required**
Engine	**SHUT DOWN and FEATHER before seizure if altitude and terrain allow (Engine Failure in Flight)**

Feather while oil pressure remains – the counter-weighted propeller needs oil pressure to move to feather before the start locks engage.

Deferred: Low oil pressure – precautionary engine shutdown; single-engine landing, land as soon as possible.	**NOTED**

No – gauge suspect:

Flight	**CONTINUE, monitor oil temperature and pressure**
Deferred: Oil pressure indication suspect – maintenance check required.	**NOTED**

Fuel Flow Fluctuation / Vapour / Rough Running (one engine)

Affected engine auxiliary fuel pump	**LOW**
Affected engine mixture	**ADJUST for smooth running**
Fuel selector	**CHECK MAIN (not AUX) – feel for detent**
Affected engine magnetos	**CHECK L / R individually, then BOTH**
Engine instruments	**CHECK (CHT, fuel flow, oil pressure)**

▶ Fuel flow steady and engine smooth?

Yes:

Deferred: Fuel flow fluctuation cleared – leave aux pump LOW; report to maintenance.	**NOTED**

No – fuel flow very low (engine-driven pump failed):

Affected engine auxiliary fuel pump	**HIGH (reduces to low pressure automatically at low throttle settings)**
Affected engine mixture	**LEAN as required for smooth running (HIGH pump floods at low power)**
Deferred: Engine-driven fuel pump suspect – aux pump HIGH, land as soon as practical; expect rough running at low power.	**NOTED**

No – still rough:

Affected engine power	**REDUCE to smoothest setting – feather if power unusable (Engine Failure in Flight)**
Deferred: Engine rough – land as soon as practical at nearest suitable airport.	**NOTED**

Crossfeed Operation / Fuel Imbalance

One engine inoperative or main tank imbalance

Flight attitude ..... **LEVEL (crossfeed in level flight only)**

Operating engine fuel selector ..... **CROSSFEED (opposite MAIN tank) – feel for detent**

Fuel flow ..... **CHECK steady after selection**

Auxiliary fuel pump (operating engine) ..... **LOW if fuel flow fluctuates**

Only the MAIN (tip) tanks can be crossfed. Aux and locker fuel on the dead-engine side is unusable except by transfer to that side's main tank (locker) – aux fuel is lost.

Fuel quantity ..... **MONITOR – return to own MAIN before descent, approach and landing**

Deferred: Crossfeed used – selectors back to own MAIN for landing; fuel on dead-engine side partly unusable. ..... **NOTED**

Propeller Overspeed

RPM above 2700 and not responding to prop lever

Affected engine throttle ..... **REDUCE**

Affected engine propeller ..... **LOW RPM (toward feather detent)**

Airspeed ..... **REDUCE to bring RPM within limits**

Affected engine oil pressure ..... **CHECK (governor is oil-operated)**

▶ RPM controllable at or below 2700?

Yes:

Power ..... **SET minimum that keeps RPM in limits**

Deferred: Propeller governor fault – control RPM with throttle and airspeed, land as soon as practical. ..... **NOTED**

No:

Affected engine ..... **SHUT DOWN and FEATHER (Engine Failure in Flight)**

Deferred: Engine shut down for prop overspeed – single-engine landing, land as soon as possible. ..... **NOTED**

Flap Failure / Asymmetry

Flap handle ..... **NEUTRAL / stop – do not continue extension**

Flap circuit breaker ..... **CHECK / RESET once**

▶ Flaps operable and symmetric?

Yes:

Flap setting ..... **AS REQUIRED**

No:

Aileron / rudder ..... **AS REQUIRED to hold wings level; airspeed above 100 KIAS**

Landing ..... **AT CURRENT FLAP SETTING – add 10 KIAS to approach speed (flaps up approx 100–105 KIAS), longer runway**

Deferred: Flaps inoperative / asymmetric – flaps-up or partial-flap landing, increased landing distance. ..... **NOTED**

Landing Gear Unsafe / Fails to Retract or Extend Normally

Red gear unsafe light, no green light, or warning horn

Airspeed ..... **BELOW 140 KIAS**

Gear light bulbs / navigation light dimming ..... **CHECK (lights dim with nav lights on)**

Landing gear circuit breaker ..... **CHECK / RESET once**

Landing gear handle ..... **CYCLE once to the desired position**

▶ Gear in desired position?

Yes:

Gear indication ..... **CONFIRM green, unsafe light out**

Deferred: Gear system fault – do not retract after landing; maintenance inspection required. ..... **NOTED**

No – failed to retract:

Landing gear handle ..... **DOWN – leave down**

Airspeed ..... **BELOW 140 KIAS**

Deferred: Gear will not retract – gear down, reduced performance and range (negative single-engine climb), land as soon as practical. ..... **NOTED**

No – failed to extend:

Use Landing Gear Emergency Extension (Hand Crank).

Cabin or Nose Baggage Door Open in Flight

Aircraft ..... **CONTROLLABLE – fly the airplane**

Airspeed ..... **REDUCE to approx 120 KIAS (or less if buffet)**

Belts / loose items ..... **SECURE**

Do not attempt to close a door in flight. An open nose baggage door can affect pitch and airflow to the engines – land as soon as possible, avoid abrupt manoeuvres and sideslips.

Deferred: Door unlatched – return and land, normal approach speed with slightly higher power; expect noise and buffet. ..... **NOTED**

Gyro Suction / Attitude Indicator Failure

Suction gauge	**CHECK (red pump indicator shows which pump failed – remaining pump maintains system)**
Attitude / heading indicators	**CROSS-CHECK with turn coordinator, compass, GPS, standby instruments**
Failed instruments	**COVER**
Autopilot	**OFF if coupled to failed gyro**
IMC	**EXIT / AVOID if both pumps or the attitude indicator failed**
Deferred: Gyro system failure – partial panel; prefer VMC approach.	**NOTED**

Inadvertent Icing

Pitot heat	**ON**
Propeller de-ice / windshield de-ice / surface de-ice boots	**ON (if installed)**
Cabin heat / windshield defrost	**ON**
Alternate induction air	**AUTOMATIC – expect a drop in manifold pressure**
Course / altitude	**CHANGE to leave icing conditions**
Power	**INCREASE to maintain airspeed**

▶ Ice remaining on airframe for landing?

No:

Yes:

Approach	**FLAPS 15° or less, increase approach speed by 10–15 KIAS, no slips**
Deferred: Residual airframe ice – partial-flap landing, approach 10–15 KIAS faster, longer runway, higher stall speed.	**NOTED**

Pitot / Static Blockage

Erratic or frozen airspeed, altimeter or VSI

Pitot heat	**ON**
Alternate static source	**ON (pull) if static blockage suspected**
Cabin vents / heater	**CLOSED (alternate static reads cabin pressure)**
Airspeed	**FLY ATTITUDE and POWER; use GPS ground speed as a cross-check**
Deferred: Pitot-static fault – airspeed / altitude unreliable, fly known power and attitude on approach, allow altimeter error on alternate static.	**NOTED**

Emergency procedures

Engine Failure During Takeoff

On the runway or just airborne with runway remaining

▶ Airborne?

No – below approx 90 KIAS:

Throttles (memory)	**CLOSED (both)**
Brakes (memory)	**MAXIMUM – stop straight ahead**
Mixtures	**IDLE CUTOFF**
Fuel selectors	**OFF**
Magneto switches	**OFF**
Battery & alternator switches	**OFF**

Yes – runway remaining:

Throttles (memory)	**CLOSED (both)**
Land (memory)	**STRAIGHT AHEAD, gear down**
Brakes	**MAXIMUM**
Mixtures / fuel selectors / magnetos	**OFF after stop**

Yes – no runway remaining:

Go to Engine Failure After Liftoff.

Below Vmca (75 KIAS) directional control cannot be maintained with full power on one engine. Do not attempt to continue a takeoff below approx 90 KIAS.

Engine Failure After Liftoff

Airborne, insufficient runway to land ahead

LAND AS SOON AS POSSIBLE

- Mixtures (memory) ..... **FULL RICH**
- Propellers (memory) ..... **HIGH RPM**
- Throttles (memory) ..... **FULL**
- Flaps (memory) ..... **UP**
- Airspeed (memory) ..... **106 KIAS (Vyse); 93 KIAS (Vxse) only to clear obstacles – never below 75 KIAS**

▶ Landing gear position?

Gear DOWN:

With gear down and the propeller windmilling, single-engine climb is negative. Retract if committed to fly; otherwise land straight ahead.

- Landing gear (memory) ..... **UP (if committed to fly)**
- Inoperative engine (memory) ..... **IDENTIFY – dead foot, dead engine**
- Inoperative engine throttle (memory) ..... **VERIFY – retard, no change**
- Inoperative engine propeller (memory) ..... **FEATHER (before RPM falls below approx 800 – start locks)**
- Inoperative engine mixture (memory) ..... **IDLE CUTOFF**

Gear UP:

- Inoperative engine (memory) ..... **IDENTIFY – dead foot, dead engine**
- Inoperative engine throttle (memory) ..... **VERIFY – retard, no change**
- Inoperative engine propeller (memory) ..... **FEATHER (before RPM falls below approx 800 – start locks)**
- Inoperative engine mixture (memory) ..... **IDLE CUTOFF**

- Bank ..... **3–5° toward operating engine, ball half out**
- Trim ..... **RUDDER & AILERON for straight flight**
- Climb ..... **106 KIAS to safe altitude, then Engine Securing**

Deferred: One engine inoperative after takeoff – single-engine approach and landing, no go-around once committed. ..... **NOTED**

Engine Failure in Flight

Loss of power on one engine at altitude

LAND AS SOON AS POSSIBLE

- Mixtures (memory) ..... **FULL RICH**
- Propellers (memory) ..... **HIGH RPM**
- Throttles (memory) ..... **FULL (as required)**
- Landing gear / flaps (memory) ..... **UP**
- Airspeed (memory) ..... **106 KIAS (Vyse) minimum**
- Inoperative engine (memory) ..... **IDENTIFY – dead foot, dead engine**
- Inoperative engine throttle (memory) ..... **VERIFY – retard, no change**
- Inoperative engine propeller (memory) ..... **FEATHER**
- Inoperative engine mixture (memory) ..... **IDLE CUTOFF**

- Bank / rudder ..... **3–5° toward operating engine, ball half out**
- Trim ..... **RUDDER & AILERON**

▶ Cause obvious (fuel selector / tank) and altitude available for a restart?

Yes – attempt air start:

- Fuel selector ..... **MAIN (if on AUX or an empty tank)**

Use the Air Start procedure. If unsuccessful, secure the engine.

No – secure engine:

Continue with Engine Securing, then divert to the nearest suitable airport.

Deferred: One engine inoperative – single-engine approach and landing, drift-down altitude check, no go-around once committed. ..... **NOTED**

Engine Securing (after feather)

Inoperative engine propeller feathered

Inoperative engine fuel selector	**OFF**
Inoperative engine auxiliary fuel pump	**OFF**
Inoperative engine magneto switches	**OFF**
Inoperative engine alternator switch	**OFF**
Inoperative engine cowl flap	**CLOSED**
Wing locker transfer pump (that side)	**OFF**
Propeller synchrophaser	**OFF (if fitted)**
Electrical load	**REDUCE to within one alternator (100 A)**
Operating engine	**MONITOR – power as required, cowl flap as required, mixture adjusted**
Fuel crossfeed	**AS REQUIRED – operating engine selector to opposite MAIN, level flight only (aux tanks cannot be crossfed)**
Trim	**ADJUST for straight flight**
ATC	**ADVISE – nearest suitable airport**

Air Start (Unfeather)

Restart attempt with altitude and cause identified

Fuel selector	**MAIN**
Auxiliary fuel pump	**LOW**
Throttle	**1/4 OPEN**
Mixture	**FULL RICH**
Magneto switches	**ON (both)**
Propeller	**FORWARD of feather detent to mid-range (unfeathering accumulator, if fitted, spins the prop)**
Starter	**ENGAGE until propeller windmills (if no accumulator)**

▶ Engine restarted?

Yes:

Throttle	**1500 RPM until oil pressure / temperature normal, then as required**
Alternator switch	**ON**
Auxiliary fuel pump	**OFF**
Mixture / cowl flap	**AS REQUIRED**
Deferred: Engine restarted in flight – land as soon as practical, cause investigation required.	**NOTED**

No:

Propeller	**FEATHER**
Mixture	**IDLE CUTOFF**

Complete Engine Securing.

Single-Engine Approach, Landing & Go-Around

One engine secured

Operating engine fuel selector	**MAIN (crossfeed as required, back to own MAIN for landing)**
Operating engine auxiliary fuel pump	**LOW**
Mixture	**FULL RICH**
Propeller	**HIGH RPM**
Cowl flap	**AS REQUIRED**
Airspeed	**106 KIAS (Vyse) until landing assured**
Approach	**NORMAL glidepath or slightly high – avoid low, flat approaches**
Landing gear	**DOWN when landing assured (within gliding distance, below 140 KIAS)**
Flaps	**15° when landing assured; FULL on short final only**
Airspeed	**REDUCE below 106 KIAS only on short final (90–95 KIAS)**

▶ Go-around required?

No:

Landing	**NORMAL – brakes as required**

Yes – above decision height:

Operating engine	**FULL POWER**
Flaps	**UP**
Landing gear	**UP (positive climb)**
Airspeed	**106 KIAS (Vyse)**
Bank	**3–5° toward operating engine**

A single-engine go-around below approx 50 ft AGL, or once full flaps are extended, should not be attempted.

Engine Fire in Flight

LAND IMMEDIATELY

- Affected engine fuel selector (memory) ..... **OFF**
- Affected engine mixture (memory) ..... **IDLE CUTOFF**
- Affected engine propeller (memory) ..... **FEATHER**
- Affected engine auxiliary fuel pump (memory) ..... **OFF**
- Affected engine throttle ..... **CLOSED**
- Affected engine magneto switches ..... **OFF**
- Affected engine alternator switch ..... **OFF**
- Affected engine cowl flap ..... **CLOSED**
- Wing locker transfer pump (that side) ..... **OFF**
- Cabin heater ..... **OFF**
- Airspeed ..... **106 KIAS or higher to help extinguish fire**
- Landing ..... **NEAREST SUITABLE AIRPORT – do not attempt restart**

Engine Fire During Start

- Starter (memory) ..... **CONTINUE CRANKING**

▶ Engine starts?

Yes:

- Throttle ..... **1700 RPM for a few minutes**
- Engine ..... **SHUT DOWN and inspect for damage**

No:

- Mixture ..... **IDLE CUTOFF**
- Throttle ..... **FULL OPEN**
- Auxiliary fuel pump ..... **OFF**
- Fuel selector ..... **OFF**
- Starter ..... **CONTINUE CRANKING to draw fire into engine**
- Magnetos / battery / alternators ..... **OFF**
- Fire extinguisher ..... **OBTAIN / USE**
- Aircraft ..... **EVACUATE**

Cabin / Electrical Fire

LAND AS SOON AS POSSIBLE

- Battery & alternator switches (memory) ..... **OFF**
- Avionics power switch (memory) ..... **OFF**
- Cabin heater (memory) ..... **OFF**
- Cabin vents ..... **CLOSED**
- Fire extinguisher ..... **ACTIVATE (then ventilate cabin)**

▶ Fire out and electrical power required?

Yes:

- Circuit breakers ..... **CHECK – do not reset faulted**
- Battery switch ..... **ON**
- Alternators / avionics / equipment ..... **ON one at a time, check for smoke**
- Deferred: Electrical fire – suspect circuits isolated, affected systems unavailable. ..... **NOTED**

No:

Landing gear and flaps are electric: plan a hand-crank gear extension and a flaps-up landing.

- Deferred: Electrical power off – hand-crank gear extension, flaps-up landing, no landing lights or gear lights. ..... **NOTED**

Emergency Descent

- Throttles (memory) ..... **CLOSED**
- Propellers (memory) ..... **HIGH RPM**
- Landing gear (memory) ..... **DOWN (below 140 KIAS)**
- Flaps (memory) ..... **15° (below 160 KIAS)**
- Airspeed (memory) ..... **140 KIAS**
- Mixtures ..... **ENRICH as required**
- Cowl flaps ..... **CLOSED**

Landing Gear Emergency Extension (Hand Crank)

Gear does not extend electrically

Airspeed	**140 KIAS or less (slower eases cranking)**
Landing gear circuit breaker	**PULL**
Landing gear handle	**DOWN**
Emergency crank	**REMOVE cover and ENGAGE (floor between front seats)**
Emergency crank	**TURN to the stop (approx 50 turns) until gear down light illuminates**
Landing gear lights	**GREEN – gear down, unsafe light out**
Emergency crank	**LEAVE engaged / stow after landing**

Do not reset the gear circuit breaker or move the gear handle in flight after a manual extension. Leave the gear down.

▶ Green light obtained?

Yes:

Deferred: Gear extended by hand crank – do not retract; gear system inspection required. ..... **NOTED**

No:

Continue to Gear-Up / Partial Gear Landing procedure.

Gear-Up / Partial Gear Landing

Gear cannot be confirmed down

Runway	**LONGEST HARD SURFACE available, emergency services advised**
Passenger briefing	**COMPLETE (belts tight, brace, exits)**
Landing gear	**UP (or as it is – do not cycle)**
Flaps	**FULL**
Approach	**NORMAL, 90–95 KIAS**
Auxiliary fuel pumps	**OFF**
Cabin door	**UNLATCHED**

On short final:

Mixtures	**IDLE CUTOFF**
Fuel selectors	**OFF**
Magneto switches	**OFF**
Battery & alternator switches	**OFF**
Touchdown	**MINIMUM SPEED, slightly nose high, wings level**
Aircraft	**EVACUATE once stopped**

Inadvertent Spin

Intentional spins prohibited – most likely during a single-engine stall or Vmca demonstration

Throttles (memory)	**CLOSED (both)**
Rudder (memory)	**FULL OPPOSITE to rotation**
Ailerons (memory)	**NEUTRAL**
Control wheel (memory)	**BRISKLY FORWARD**
Rudder	**NEUTRALISE when rotation stops**
Recovery	**SMOOTH pull-out from dive, power as required**

Tip tank fuel increases spin inertia; recovery may take several turns. Practise single-engine work and Vmca demonstrations at a safe altitude only, and recover at the first sign of stall or loss of directional control.

Prototype content. Verify against the operator's approved checklist and the POH/AFM before any use. AeroE Checklist 7f510494 · 2026-11-07